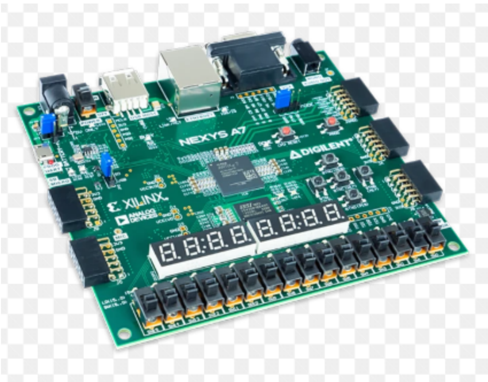
HARDWARE ENGINEERING LAB
Hamm-Lippstadt University of Applied Sciences (HSHL)

Password-Protected Parking System Controller

VHDL Design and Implementation on the Digilent Nexys A7-100T FPGA



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Abstract

This document presents the design, implementation, and verification of a password-protected parking system controller developed entirely in VHDL and deployed on a Digilent Nexys A7-100T FPGA board. The controller enforces access control through a 4-bit switch-entered password, tracks real-time parking occupancy against a 15-space capacity limit, and reports system status through a multiplexed seven-segment display and four status LEDs. The design is partitioned into two synchronous VHDL entities: a reusable debounce module that conditions all mechanical button inputs, and a top-level controller that performs password verification, occupancy counting, and display multiplexing. The implementation was validated across three stages — functional simulation of reset, password-verification, and entry/exit sequencing behavior; synthesis and implementation targeting the Nexys A7-100T under the constraints listed in Section 5; and hardware testing on physical FPGA hardware, confirmed through the board photographs in Section 6. A companion custom PCB, targeting a Lattice MachXO2-640HC FPGA in place of the Nexys A7 prototyping board, was additionally developed in Altium Designer and is documented in Section 7 through complete schematic capture, a 3D assembly render, and the final routed copper layout. Together, these results demonstrate a complete and verified design path from behavioral VHDL description through to a manufacturable embedded access-control board.

Keywords: FPGA, VHDL, access control, seven-segment display, switch debouncing, Altium Designer, PCB design, embedded systems

1 Project Overview

This project implements a password-protected parking system controller in VHDL, designed and tested on an FPGA board. The system controls vehicle entry and exit, maintains a running count of available parking spaces, and prevents new entries once the lot is full.

The parking lot has a maximum capacity of 15 spaces. The controller maintains an internal count of occupied spaces and computes the number of available spaces as:

$$\text{available spaces} = 15 - \text{occupied spaces}.$$

At a glance
Capacity: 15 spaces | Password: 4-bit (SW[3:0]) | Clock: 100 MHz
Debounce: 20 ms | Display: 4-digit 7-segment | Target: Xilinx Artix-7 (Nexys A7-100T)

The available space count is shown on a seven-segment display. The two rightmost digits display the number of free spaces, while the leftmost digit indicates password verification status:

- **A** - password accepted
- **E** - wrong password entered

2 Access Control

Vehicle entry is protected by a simple password mechanism. The password is entered using four on-board switches, which simulate an RFID/card-based access input.

2.1 Authorized Code

In the VHDL source, the authorized code is defined as a 4-bit constant compared directly against the switch vector `SW(3 downto 0)`:

```
constant AUTHORIZED_CODE : STD_LOGIC_VECTOR(3 downto 0) := "1010";
```

This means the required logic levels are SW3 = 1, SW2 = 0, SW1 = 1, SW0 = 0. On the physical Nexys A7 board, switch SW0 is the rightmost switch on the panel; physically setting switches SW0–SW3 to match this pattern (SW0 down, SW1 up, SW2 down, SW3 up) satisfies the comparison in hardware.

2.2 Verification Procedure

Before a vehicle may enter, the user must:

1. Set the four switches SW[3:0] to the access code.
2. Press BTNL (Verify Password).

If the code is correct and the lot is not full, the display shows **A** and one vehicle is permitted to enter on the next press of BTNU (Entry). If the code is incorrect, the display shows **E** and entry is blocked until verification succeeds.

3 Controls and Indicators

User interaction with the parking system controller is handled entirely through the Nexys A7's on-board push buttons, switches, LEDs, and seven-segment display.

3.1 Push Buttons

Four push buttons drive the system's core actions: resetting the controller, verifying the entered password, and registering vehicle entry or exit events.

Button	Function
BTNC	Reset
BTNL	Verify password
BTNU	Entry
BTND	Exit

Table 1: Push button assignments

3.2 LED Status Indicators

Four LEDs give an at-a-glance summary of parking availability and password verification status. LED[15:4] are unused and held low.

LED	Indicates
LED0	Space available
LED1	Parking full
LED2	Password accepted
LED3	Password rejected

Table 2: LED status assignments (LED[15:4] unused, held low)

3.3 Debounce Logic

All four push buttons are routed through a dedicated debounce pulse module before being used by the control logic, in order to suppress false multiple-trigger events caused by mechanical switch bounce. Each instance double-synchronizes the input and requires the signal to remain stable for a configurable number of clock cycles (`STABLE_COUNT`, default 2,000,000 cycles at 100 MHz $\approx$ 20 ms) before registering a clean single-cycle pulse.

4 System Architecture

The design is partitioned into two VHDL entities:

- `debounce_pulse` — a reusable synchronizer/debounce module instantiated four times (entry, exit, reset, verify buttons).
- `parking_top` — the top-level entity containing the password comparison logic, occupancy counter, seven-segment multiplexing, and LED status outputs.

The four password switches connect directly to `parking_top`, since switches remain electrically stable and do not require debouncing. Both entities operate from the same 100 MHz FPGA clock, keeping all logic synchronized.

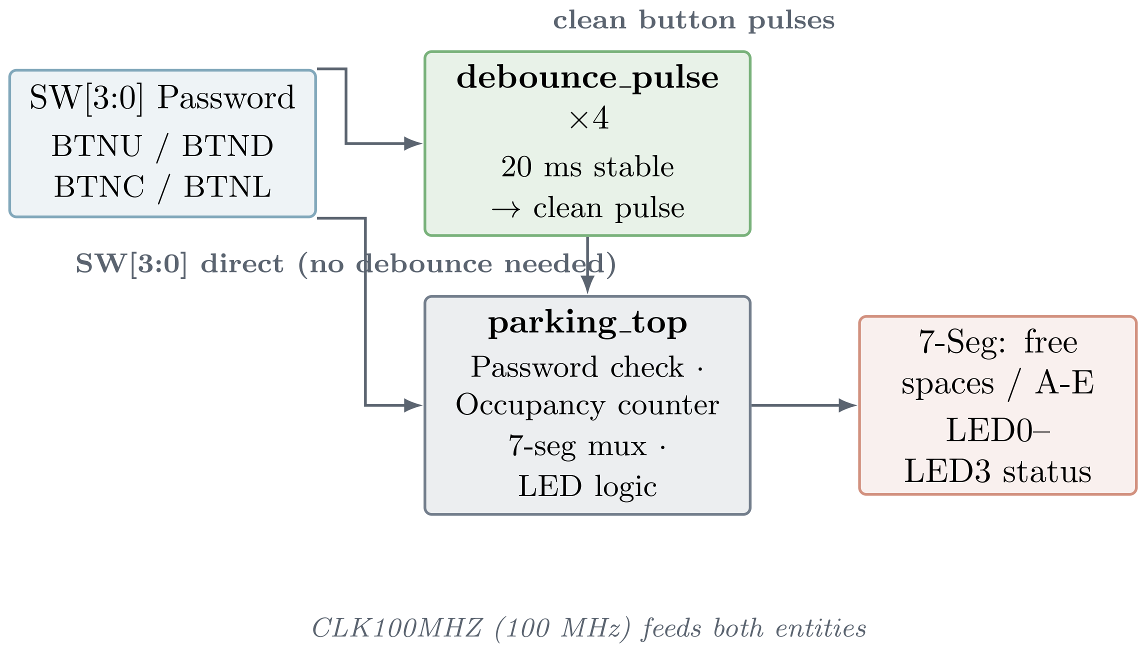

Figure 1: System architecture: the debounce module and top-level controller entity, and their I/O connections

4.1 Occupancy and Display Logic

The occupancy counter increments on a valid, password-authorized entry pulse and decrements on an exit pulse, saturating at 0 and 15. The available count is split into tens and ones digits and multiplexed across the seven-segment display anodes using a refresh counter, cycling through three display positions: ones digit, tens digit, and the access-status character (A/E).

4.2 Source Code

`debounce_pulse`:

```
entity debounce_pulse is
  generic ( STABLE_COUNT : natural := 2000000 );
  Port ( clk : in STD_LOGIC;
         btn_in : in STD_LOGIC;
         pulse : out STD_LOGIC );
end debounce_pulse;
```

`parking_top` (port interface):

```
entity parking_top is
  generic ( DB_COUNT : natural := 2000000 );
  Port ( CLK100MHZ : in STD_LOGIC;
         BTNU : in STD_LOGIC; -- Entry
         BTND : in STD_LOGIC; -- Exit
         BTNC : in STD_LOGIC; -- Reset
         BTNL : in STD_LOGIC; -- Verify password
         SW : in STD_LOGIC_VECTOR(3 downto 0); -- Password input
         LED : out STD_LOGIC_VECTOR(15 downto 0);
         CA, CB, CC, CD, CE, CF, CG, DP : out STD_LOGIC;
         AN : out STD_LOGIC_VECTOR(7 downto 0) );
end parking_top;
```

Password verification and the occupancy counter are the two decision points that most directly affect correctness:

```
-- Password check (on BTNL press)
elsif verify_p = '1' then
  if SW = AUTHORIZED_CODE then
    pw_ok <= '1'; -- grant access
  else
    pw_ok <= '0'; -- block entry
  end if;

-- Entry / exit counter
if entry_p='1' and pw_ok='1' and occupied<15 then
  occupied <= occupied + 1;
elsif exit_p='1' and occupied>0 then
  occupied <= occupied - 1;
end if;
```

Full source listings for both entities, including the seven-segment encoding functions, are included in the project repository under `src/`.

5 Pin Assignments (Nexys A7-100T XDC)

The following tables list the physical FPGA pin assignments used in the project's constraints (`.xdc`) file, grouped by function.

Signal	Pin	Standard / Function
CLK100MHZ	E3	LVCMOS33, 100 MHz
SW[0]	J15	LVCMOS33
SW[1]	L16	LVCMOS33
SW[2]	M13	LVCMOS33
SW[3]	R15	LVCMOS33
BTNU	M18	Entry
BTND	P18	Exit
BTNC	N17	Reset
BTNL	P17	Verify password

Table 3: Clock, password switch, and push button pin assignments

Of the sixteen available board LEDs, only LED[0]–LED[3] are functionally used by the design (Section 3.2); the remaining twelve are reserved and held low. The seven-segment display uses eight cathode/decimal-point signals (CA–CG, DP, mapped to pins T10, R10, K16, K13, P15, T11, L18, H15) and eight anode select lines (AN0–AN7, with AN0 as the ones digit, AN1 as the tens digit, and AN7 as the status character).

6 Verification and Testing

The design was verified using the following flow:

1. **Functional simulation** — a VHDL testbench was used to exercise reset, password verification (correct and incorrect codes), entry/exit sequencing, and full-lot behavior.
2. **Synthesis and implementation** — the design was synthesized and implemented in Vivado targeting the Nexys A7-100T using the constraints in Section 5.
3. **Hardware testing** — the bitstream was loaded onto physical FPGA hardware and verified against the button, switch, LED, and seven-segment behavior described above.

Fig. 2 shows the controller running on the physical Nexys A7-100T board during hardware testing: an incorrect password (E, 12 free spaces), an accepted password (A, 15 free spaces), and the board immediately after reset (15 free spaces, no status character).

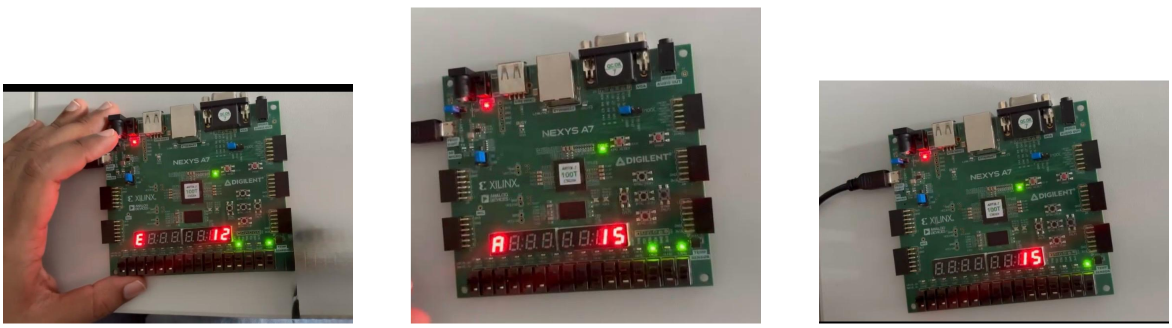

Figure 2: Hardware verification on the Nexys A7-100T: incorrect password (left), accepted password (center), state immediately after reset (right)

7 Hardware Concept (PCB)

A custom PCB implementation of this design was developed in Altium Designer, targeting a Lattice MachXO2-640HC FPGA in place of the Nexys A7 prototyping board. The planned board includes password input switches and push buttons, status LEDs, a seven-segment display, FPGA interface connections, power supply circuitry, a clock source, and configuration/programming circuitry.

7.1 FPGA I/O Banks and Power Schematic

Fig. 3 shows the schematic sheet for the FPGA itself, organized by I/O bank: Bank 0 carries the clock and status LEDs, Bank 1 the password switches and push buttons, Bank 2 the seven-segment cathode segments, and Bank 3 the anode select lines, together with the 3.3 V power and decoupling network.

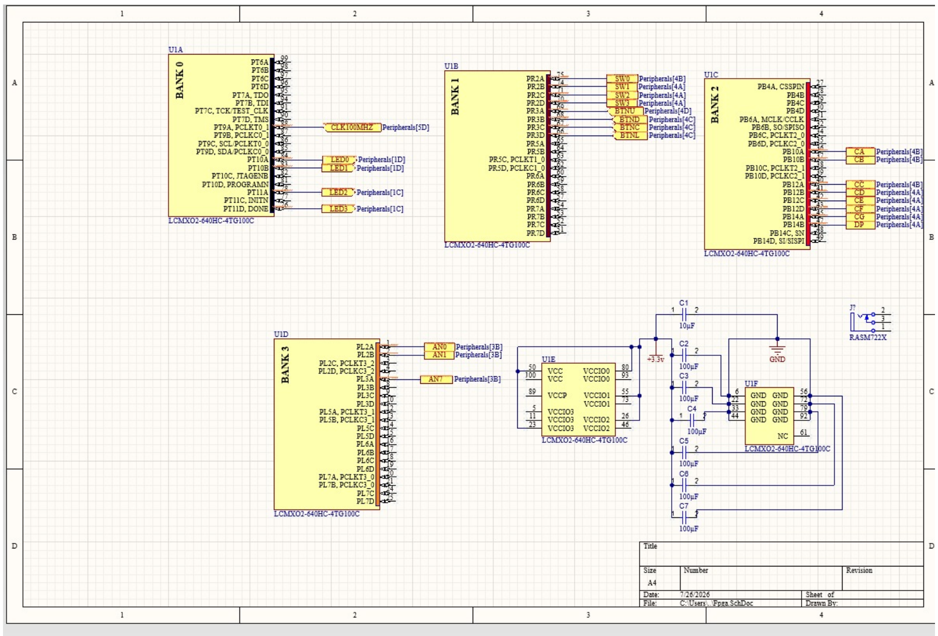

Figure 3: FPGA I/O banks and power supply schematic

7.2 Parking System Schematic

Fig. 4 shows the peripheral-side schematic: push buttons, the password DIP switch, status LEDs, and the seven-segment display, each wired through appropriate resistor networks to the FPGA I/O pins.

7.3 PCB Design and Routed Layout

Fig. 5 shows Altium Designer's 3D render of the assembled board alongside the final routed copper layout, which forms the direct input to the manufacturing (Gerber) files used for board fabrication.

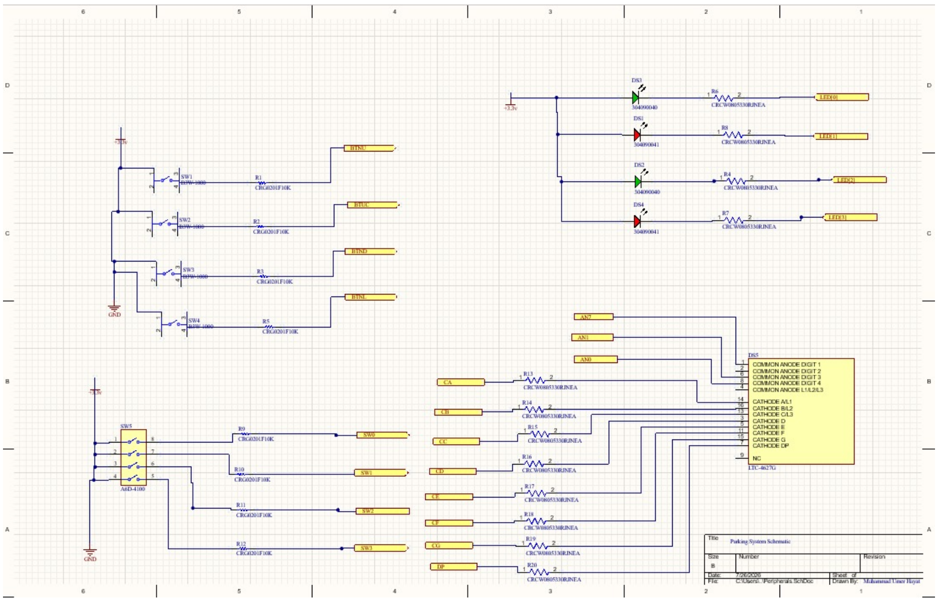

Figure 4: Parking system schematic: buttons, switches, LEDs, and seven-segment display

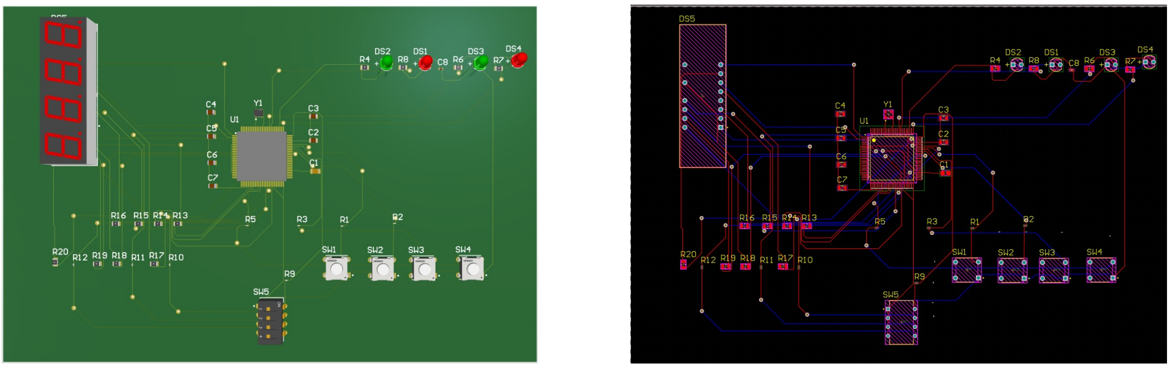

Figure 5: PCB design: 3D render of the assembled board (left) and routed PCB layout in Altium Designer (right)

8 Conclusion

This document presented a password-protected parking system controller implemented in VHDL and verified on a Nexys A7-100T FPGA, covering access control, occupancy tracking, seven-segment/LED status reporting, and switch debouncing. The design was validated through simulation, synthesis, and physical hardware testing, and a companion custom PCB targeting a Lattice MachXO2-640HC FPGA was developed in Altium Designer, including schematic capture, 3D visualization, and routed board layout. Together, these results demonstrate a complete, verified design path from behavioral VHDL description to a manufacturable embedded access-control board.

References

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